AI ETHICS

Trust in Artificial Intelligence

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Building trust in AI systems and technologies is critical for our future. Drawing on my extensive research into the subject, I will highlight key insights from our multicountry survey to answer questions ranging from whether people trust in AI systems, to the expectations of its governance and use, and what can be done to support community trust in AI. Our research also highlights several pathways to enhancing trust and acceptance of AI systems. I will conclude this module with an introduction to the business model and practical guidelines on achieving trustworthy AI, which we developed as part of our research.

Lecture Transcript

0:03 Hello, everyone. I'm Nicole Gillespie. I'm a professor of Management and KPMG Chair of Organizational Trust at the University of Queensland in Australia. I'm a fellow of the Academy of Social Sciences, and my training is in Organizational Psychology and Management. So over the past 20 years, I've been researching trust in organizational contexts. And in particular, in the last few years, I focused in on trust in emerging technology and technology mediated services. So I'm currently leading two programs of research that are relevant to artificial intelligence. The first examines community trust and attitudes towards AI and its governance and management. And the second examines how organizations can develop and use AI in a trustworthy and responsible way. So over the next 20 minutes or so, I'm going to focus on sharing insights from our recent survey of over 6000 people in five countries examining community trust and attitudes towards AI. In doing so, I'll provide some insights to the following questions, ranging from whether people trust in AI systems, to the expectations of its governance and use, and what can be done to support community trust in AI. I'll also share our research examining how organizations can go about achieving trustworthy AI and our model to guide organizations on that journey. So let's start by asking, why are we focused on trust in AI? Well, there's at least four reasons here. First, AI is a cornerstone technology of the Fourth Industrial Revolution, described by Klaus Schwab, the chairman of the World Economic Forum, as a fusion of technologies that is blurring the lines between the physical, digital and biological spheres. Now, this fusion promises new solutions to 21st century challenges related to areas such as health care, climate change, agricultural sustainability and productivity. And AI is involved in this through sophisticated solutions such as precision medicine, autonomous vehicles, digital twins, and augmented humans. There are opportunities to revolutionize our workplaces and our urban environments. AI is increasingly a ubiquitous part of our everyday lives and is already transforming the way that we live and work. From the smart speakers in our home, our traffic navigation apps, social media, ride sharing, email filters - all of these use AI. And increasingly, these technologies are being used in our workplaces and organizations to enable better predictions and decisions to support intelligent automation and advanced data analytics. This is being applied across the supply chain right through to the customer and employee experience. For example, we're seeing better fraud detection in banks through AI. It is being used to enhance monitoring of crop and soil health in agriculture. And it's also being used to identify modern slavery and supply chains, and even modeling the spread of COVID-19 to inform government decision making. All sectors of the global economy are now embracing AI. And there's been an exponential increase in investment over the past decade, and this has only been accelerated through the pandemic.

3:22 Now, while the benefits and promise of AI for society are undeniable, so too are the very real risks and challenges AI poses. These include cases of AI codifying and reinforcing biases, resulting in discrimination in hiring, parole decisions and access to medical care. We're seeing breaches of privacy, the spreading of fake and manipulative online content. There are concerns about technological unemployment, mass surveillance undermining human rights through facial recognition technology. There's the loss of human agency, which is a fear - a fear of dehumanization. This concerns critical AI failures, as well as lethal autonomous weapons, and it's the most vulnerable in our population. So they are typically the ones most at risk of harm, and are least able to access remedy to such problems. So our review of the literature highlights four key AI trust challenges, and these are centered around: explainability and transparency, accuracy and bias, human agency and control given AI's self learning and automation capability, as well as data privacy and security. So these issues are causing much concern and raising questions about the trustworthiness of these AI systems. Yet trust is critical to innovation and acceptance of AI systems. The benefits and promise of AI will be much harder to realize, slower to achieve and more expensive if people do not trust in the technology. Trust is also critical to reputation. We've already seen organizations suffering negative reputational damage due to trust breaches involving AI. Overtrust in AI is also well documented. Some people have unrealistic expectations of AI capabilities and override their own judgment and experience, even when it's more accurate. Overtrusting complacency such as this opens up risk of harm. But at the same time, algorithmic aversion and unwarranted distrust also prevents the realization of the many benefits of AI for society. So well calibrated trust is our goal here. Now, despite the central importance of trust, today, little is known about people's trust in AI and what influences it. So this inspired us to go about conducting a multi-country survey with the aim of understanding people's trust in AI and their broader attitudes and expectations around its development, use and governance. We wanted to explore similarities and differences across countries, given AI systems are not bounded by physical borders and are being rapidly deployed across the globe. We also want to enable benchmarking of changes in trust and attitudes over time. We'll be conducting this survey biannually and extending our data collection to be truly global in 2022. I'd like to acknowledge my wonderful colleagues who have helped with this project, the UQ research team, Steve Lockey, and Caitlin Curtis and myself, and we've been facilitated by an expert panel of advisors from across KPMG. Now I'm just going to highlight a snapshot of the research findings. The report itself goes into much more depth and breadth, and it's publicly available from the University of Queensland and KPMG websites. Now, our insights are based on nationally representative samples from five western countries: the USA, Germany, Australia, Canada, and the UK. Each of these countries are rapidly investing in AI, and each has a national AI strategy. We surveyed over 6,000 people in total with a minimum of 1,200 in each country. Each country sample was nationally representative on gender, age and geographic location matched against its national census data. And here you can see the percentages for the full sample across all countries. We collected the data using research panels in late 2020. And a key insight that we found through the analyses was that there's many more commonalities than differences across countries. So I'm going to report the aggregated findings. So these are aggregated across countries, and I'll highlight any important country differences. So first, we asked Do people trust in AI? To answer this question, we asked whether people are willing to rely on and share information with AI systems. So these are the two most common ways that trust is demonstrated. We found that most people, 72% across all five countries, are wary about trusting AI systems. As shown in the dark blue bars here, about a third are unwilling to rely on the output of AI systems so their recommendations and decisions, and about two in five are unwilling to share information with an AI system. We also see over a third are ambivalent in their trust. There were no significant differences here across countries. However, we did find when we asked about trust in two specific AI applications, so these are AI applications that are in wide use and have the potential to affect many people. The first was healthcare AI used to improve the diagnosis and treatment of disease by informing clinical decision making. And the second was human resource AI, HR AI, which is used to improve the prediction of employee performance and used in hiring decisions. So we found trust depends on the AI application. People are more trusting of AI use for health care, and less trusting of AI use for human resources. So as shown by the bars on the right, only about a quarter are willing to rely on and share information with AI used for human resources, whereas over a third are willing to trust AI in the context of health care. Now the real home result for healthcare is likely due to the fact that we generally trust doctors given the purpose of healthcare is to benefit patients and their extensive training. However, we can also see that while the AI application makes a difference, trust is still low across all the applications examined. And in prior research, we actually examined a broader range of AI applications and the least trusted of these was HR AI and the most trusted was healthcare AI. So we can see that even with the most trusted application of healthcare AI, trust is still low.

9:54 We further asked about acceptance of AI and found that people generally accept or tolerate AI but few approve or embrace it. Now this is problematic as it's only when people approve or embrace AI that they're likely to engage with it and be supportive of its use. But at the same time, only a small proportion are outrightly rejecting AI. Most people are in this middle band, leaning to the side of just tolerating or just accepting it, but not approving and embracing it. We found younger people, notably generations Z and millennials are more likely to trust AI systems and approve and embrace them than older generations. But again, even these younger generations tend to report low trust and acceptance. We further wanted to understand who is trusted to develop, use and govern AI. Now we found a clear pattern across countries: people are most confident in their national university and research institutions, as well as their national defense organizations to develop and use AI. Over 70% have moderate to complete confidence in these entities. In comparison, people have less confidence in government, commercial and technology organizations. So one opportunity here is for governments and commercial companies to collaborate with more trusted entities such as universities and research institutions to develop AI. We also asked about confidence to regulate and govern AI and found a very similar pattern of findings but generally lower levels of confidence across all entities. So people again had the most confidence in universities, research institutions, as well as security and defense agencies, to govern their use of AI. And the least confidence in technology, commercial companies and government to do that. We also found some country differences, so Americans and the British are less confident in their governments to develop, use and regulate AI. This likely reflects a significantly lower general trust these two countries have in their current governance compared to the other countries that we surveyed. Now, this is supported by our finding that general trust in government is strongly correlated with specific trust in government to develop, use, regulate and govern AI. We also found that Americans are less confident in a range of entities to develop, use and govern AI compared to other countries. Furthermore, Germans were notably less confident in their defense forces and in technology companies to develop and use AI compared to the other countries. But in contrast, Australia was more confident in their universities and defense forces to develop, use and govern AI compared to other countries. Now focusing further on people's views on AI regulation, we found that the large majority of people, 81%, view AI regulation as needed. Now, this aligns with prior surveys indicating strong community desire for regulation. However, as shown by the purple and light green bars here, most people either disagree or are unsure that current regulations and laws are sufficient to make AI use safe and protect from problems. Now, the desire for stronger regulatory framework can be partly explained by the finding that 70% of people believe the impact of AI on society is uncertain and unpredictable. Digging further into this, we found that people believe a range of AI societal challenges are likely to impact large proportions of the population. So we presented descriptions of 12 AI societal challenges that have been used in a prior survey of American attitudes developed by Zhang and Dafoe. So these challenges range from AI-assisted surveillance violating privacy and civil liberties to AI spreading fake and harmful content to AI bias in HR recruiting. So each of these had a short description, and then people rated the likelihood to which they believe this would occur. So as shown in the figure, each of these AI societal challenges was viewed as more likely than unlikely to occur and impact large numbers of people. The exception was lethal autonomous weapons. Now the top four challenges, the most likely to impact society, each related to data challenges, such as surveillance, fake online content, cyberattacks and data privacy. We further found that only one in five people believe AI will create more jobs than it will eliminate, creating uncertainty about technological unemployment. Importantly, people have strong expectations that companies and governments will carefully manage each of these challenges. 85% rated this as highly important.

14:59 Now given these challenges and the strong desire for regulation, we asked who should regulate AI and the majority felt a range of entities should play a role that included a new independent AI regulator, as well as government and existing regulators. Now co-regulation and involvement of industry that uses or develops AI was also viewed as desirable by the majority. We found several differences across countries and their expectations of who should regulate AI. More British expected a new dedicated independent AI regulator than people in other countries and fewer Americans expected external regulation by government and existing regulators. However, organizations don't have to rely solely on regulations to enhance trust in their AI use. Most people indicate that they would be more willing to use an AI system if there are assurance mechanisms in place, for example, independent AI ethical reviews, or if organizations were using AI ethical codes of conduct. We also saw adherence to national standards for AI transparency and explainability and AI ethics certification can also support people to trust and use AI systems. So these assurance mechanisms signal that there's some kind of governance over the ethical and trustworthy use of AI, which is reassuring to people in the face of high perceived uncertainty. So an important goal of our research was to determine the principles and practices that are important for people to trust in AI. To answer this question, we asked about the importance of over 30 practices underlying the eight principles for trustworthy AI shown here. Now these principles were drawn from European Commission principles for trustworthy AI, and align with the AI ethical principles developed by many governments, including the five countries that we examined in this research. An example item, for example, from technical robustness and safety, was the AI system is regulated, regularly monitored and tested before and throughout the AI lifecycle to ensure it operates as intended. So our results indicate that every one of the 33 practices underlying these eight principles was deemed highly important for the public to trust in AI systems. Specifically, 95% of people endorsed each practice as important for their own trust. This provides clear public endorsement for these principles, and a blueprint for developing and using AI in a way that supports trust. Here you can see those who rated each principle and their underlying set of practices as highly important - they're the red bars - or as moderately important - that's the purple bars. And when we aggregate this up across country, you can see that 92% or more people in each country rated these principles as moderately to highly important for trust. Now, it's important to contextualize people's trust and attitudes based on their understanding. We found that most people don't feel that they understand AI or when and how it's being used. 60% report low subjective understanding of AI and we also asked about this objectively. So we presented people with 10 common applications that employ AI, such as social media and ride sharing apps, and we asked first if they had used this technology, and whether they thought it employed AI. Now the green bars show the percentage of people who are unaware that this technology uses AI. Now these findings highlight that people are often unaware that even these most common applications of AI, actually are using AI. Surprisingly, even when people reported using these applications, they still often were unaware that they employed AI. So for example, if we take social media, 76% of people report using social media. However, 59% are unaware that social media uses AI. We also found important gender and education differences here. Men and those with a university degree are more likely to understand when AI is being used, as well as report higher subjective knowledge of AI than women or those without a university education. Younger generations also reported more subjective understanding than older generations. Now, the positive sign here is that 83% of people report that they wanted to learn more about AI. So this bodes well for the uptake of any kind of literacy programs to help people to come up to speed with what AI is.

19:54 Now to understand the key drivers of trust in AI, we conducted statistical path modeling and we identified four key drivers. People are more likely to trust AI systems when they believe current regulations and laws are adequate to make AI use safe, so what we call current safeguards. They're also more likely to trust in AI systems, when they perceive AI will have a positive impact on jobs - creating more jobs, and it will eliminate. And when they have more familiarity and understanding of AI, and perceive less uncertainty about the impact of AI on society. Now, of these drivers, the perceived adequacy of current safeguards is clearly the strongest. It was over twice as predictive as the next predictor. And our modeling also confirmed what has been shown quite extensively in the literature and that is that trust strongly influences the acceptance of AI. Now in conclusion, a key insight from our survey is that trust in AI is low across all Western countries examined. Trust in our AI cannot be taken for granted. If left unaddressed, this low trust is likely to impair societal uptake and the realization of the benefits of AI. Now our findings highlight several pathways to enhancing trust and acceptance of AI systems. First, organizations need to live up to people's expectations of trustworthy AI, so community trust will only be realized and sustained if the AI systems that are being used are in fact shown to be trustworthy over the long term. So our findings reveal that people have very clear and consistent expectations of the principles and practices that they expect in terms of the design, development and deployment of AI systems. Organizational leaders can ensure that they have the right structures, processes, cultures and governance mechanisms in place to meet these expectations and make sure those principles for trustworthy AI are actually translated into practice, and that is done consistently across the AI lifecycle. The second pathway is to strengthen and communicate the framework for governing and regulating AI. People clearly don't feel that current regulations and laws are fit for purpose. Yet this is critical for community trust and acceptance. So government and regulatory bodies can work with industry to strengthen these regulatory and governance frameworks. And organizations developing and using AI can also proactively adopt best practice and mechanisms to support the trustworthy use of AI even if it's not currently mandated. A third pathway is to enhance public AI literacy as well as that of employees and consumers. Educating the community, as well as employees and consumers, about what AI is and when and how it is being used will enable people to make more informed decisions about their engagement with AI that will help them to better seize the opportunities and benefits it offers, while also identifying and managing the risks, for example around data sharing and privacy. AI literacy is also fundamental to citizen involvement in public policy and debate on a responsible stewardship of AI and other sophisticated forms of emerging technology into society and the workplace. Now to help organizations put the principles of trustworthy AI into practice, we developed a whole business model and practical guidelines in our report on achieving trustworthy AI. Now, because the benefits, challenges, risks and opportunities that AI offers differs from one industry to another, and from one application to another, it's important that organizations take a tailored approach. So our model helps organizations think through the various considerations and work practices involved in ensuring AI systems are trustworthy and meet stakeholders expectations. It highlights the need to take a holistic approach that integrates and connects the technical areas of data, algorithms and security, with the legal, ethical and management functions to ensure strong organizational alignment. Now these six dimensions need to operate in a connected way to ensure the trustworthy design, development, deployment and procurement of AI systems across the AI lifecycle. So our report includes a set of questions organizations can ask themselves to reflect on the maturity of its organizational governance and technical practices for supporting trustworthy AI and adopt the practical guidelines to strengthen any areas that are identified as being underdeveloped. So we believe that this type of proactive implementation is required to identify and prevent problems and build a sustained reputation for responsible innovation with AI.

24:48 Now all the reports that I've referred to are freely available from the University of Queensland and the KPMG websites for anyone who would like to understand and read them in more depth. The research paper I referred to is also open access. And if you'd like to learn more, please feel free to reach out through our Trust, Ethics and Governance Alliance LinkedIn site that's free to join for anyone. Thank you for your attention and interest in this work and I wish you all the very best in your learning around the ethical use of AI. Thank you.